



DIVE x WFU

04.27.2026

Campbell Charneco, Kyle Corso,
Julia Reich, Yize Sun

What core question guided this journey?

How can the DIVE team **define, measure, and optimize** the **ROI** of past and ongoing data requests?

Building a **reproducible, data-driven** ROI engine for the DIVE team

10+ interviews

*with industry professionals
and healthcare analytics
experts to ground our ROI
framework in real-world
practice*

20+ publications

*reviewed to benchmark
healthcare analytics ROI
and validate our scoring
methodology*

Scraped and transformed

*historical project requests into
analyzable data, unified
across all 9 sources into a
single clean framework*

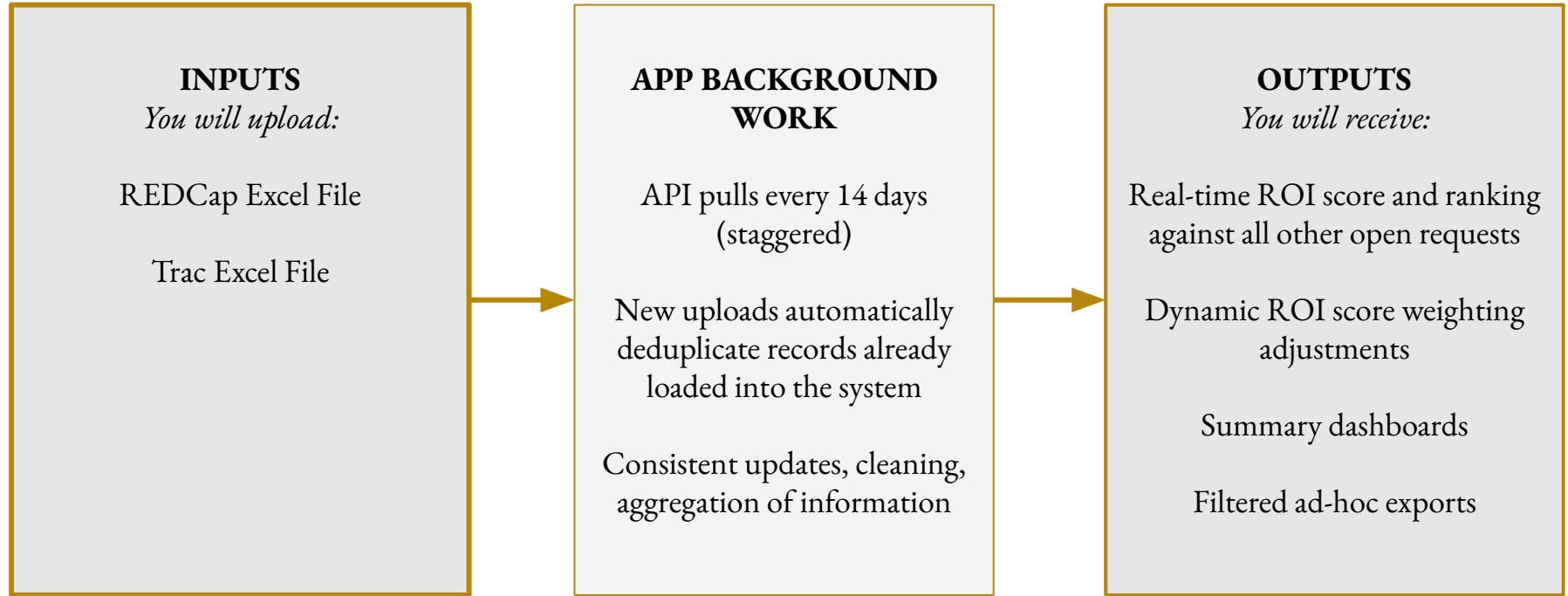
Matched project titles

*to published research and
academic citations across 9
integrated data sources*

Defined a true ROI model

*with dual-sided return and
effort scoring, anchored in a
medical specialty market
intelligence layer unique to
the DIVE team*

End user workflow enables a **streamlined ranking** of incoming requests.



Upload Data

Research Prioritizer

EMR Request ROI Scoring

- Dashboard
- PI Profiles
- Summary Stats
- Weight Config
- Admin Config
- Audit Log
- Upload Data**
- Export

Upload Data

Import REDCap and Trac Excel files. All parsing happens locally in your browser.

REDCap Export

Upload the REDCap service request export (.xlsx)



Drop .xlsx file here or click to browse
REDCap data extraction request export

Trac Hours

Upload the Trac time tracking export (.xlsx)



Drop .xlsx file here or click to browse
Trac time/hours tracking export

old_trac_filtered.xlsx

Ready to import as

Trac

× Cancel

Upload Now

Upload History

trac	old_trac_filtered.xlsx	248 records
redcap	dive_sample.xlsx	38 records
trac	datarequests_trac_filtered.xlsx	1190 records

History of which files you uploaded when appear here.

4/23/2026, 5:32:34 PM
4/23/2026, 5:32:30 PM
4/23/2026, 5:32:26 PM

Current Data

REDCap Requests: 38

Trac Hours Records: 1190

Summary of data in the tool.

Clear All Data

Remove all loaded data from browser storage. This cannot be undone.

Clear All Data

You can **clear** all uploaded data from the tool here.

Drag and drop or click to upload your **REDCap and Trac** files, respectively, one at a time. The data will not upload automatically; you will need to click the 'Upload Data' button that will appear here.

The Upload Data page is the first page you will go to. Here, you will **populate the tool with your data** from REDCap and Trac systems. If you send in duplicate requests, the tool will ignore the duplicates and only upload the new requests.

The tool is designed so your data is only **stored in your browser cache**. To clear all data from the browser, go to your browser's History page, then click "Delete browsing data."

All data stays local

Admin Configuration

Admin Configuration is your next step when you first upload data to the tool. Once saved, the settings will remain unchanged unless you change them again.

Tier 1: Project has a **full** row of data directly used to calculate ROI score.

Tier 2: **Some** data; similar projects are used as a 'proxy' to fill missing values to calculate ROI score.

Tier 3: Very **little** data; ROI score is not calculated (showing *pending*).

If you want to **discount** ROI scores that rely on similarities to other projects to 85% of its calculated ROI, toggle the Proxy Scoring ON.

Research Prioritizer

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Admin Configuration

System-wide scoring parameters. These values are used by the ROI Scoring Engine.

API Keys

Stored locally in your browser only — never sent to any server or committed to code.

NCBI API Key

.....   Save Key 

☒ API key configured — PubMed rate limit: 10 req/sec

Add **API key** for faster data extraction.

To get your API key, follow these steps:

1. While signed in to your My NCBI account, go to your [NCBI Account Settings](#). If you don't have a My NCBI account, you'll need to create one.
2. Scroll to the bottom of the page to find the "API Key Management" section.
3. Click the "Create API Key" button. A unique sequence of characters will be generated and displayed in the "API Key Management" box. This is your API key.

Proxy Scoring (Tier 2)

When enrichment data is unavailable, REDCap proxy signals are used with a confidence discount

Enable Proxy Scoring

When off, PIs without enrichment data will not receive a proxy-based score.



T2 Proxy Confidence Discount

T2 (partial-data) scores are multiplied by this value. Lower = less trust in proxy data.

0.85

T3 Insufficient-Data Discount

T3 (insufficient-data) scores are multiplied by this value. Typically lower than the T2 discount.

0.50

Effort Multiplier

Scales the weighted return based on hours worked. Anchored so that the average project length yields a neutral 1.0x (no change). Faster projects get a bonus; slower ones a penalty — dampened by the exponent and clamped to the configured band.

Anchor Hours (neutral 1.0x)

2

Projects taking this many hours yield multiplier 1.0x. Default: 2h.

Dampening Exponent

0 = disabled (always 1x), 0.5 = V-dampened (default), 1 = linear anchor/hours.

0.50

Min Multiplier (clamp)

0.25

Max Multiplier (clamp)

3

Live Preview

0.5h	1h	2h	4h	8h	20h	40h
2.00x	1.41x	1.00x	0.71x	0.50x	0.32x	0.25x

Bold cell = anchor hours (always 1.00x). Values outside [Min, Max] are clamped.

Adjust the return based on hours spent.

Include the 'investment' of time in the ROI calculation. You are able to change the settings for the adjustment if you like. Currently 2 hours means there is no change in the ROI score.

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Niche Journal Bonus

Define what counts as a "niche" journal using SCImago metrics. All enabled criteria must match (AND). When matched, the Journal Prestige subscore is multiplied by the bonus.

Enable Niche Journal Bonus

When off, no niche multiplier is applied.

Min Multiplier

1.1

Max Multiplier

1.15

SJR Quartile

Limit niche bonus to journals in selected quartiles.

☒ Q1 ☒ Q2 ☐ Q3 ☐ Q4

Numeric Metric Criteria

Total Docs (3yr)

Number of articles published in last 3 years — lower = more selective/niche

≤ (at most)

300

SJR Score

SCImago Journal Rank — higher = more prestigious

≥ (at least)

1

SJR H-index

Journal-level h-index — higher = more established

≤ (at most)

100

Citations / Doc (2yr)

Average citations per document over 2 years

≥ (at least)

2

Bonus Strength Driver

Which enabled metric scales the bonus between Min and Max. Lower values of "≤" metrics (or higher values of "≥" metrics) yield bonuses closer to Max.

Total Docs (3yr)

Active: Quartile ∈ {Q1, Q2} AND Total Docs (3yr) ≤ 300

If **Bonus** is on, any project that has an article published in a journal with these metrics will be considered 'niche,' and its ROI will increase by between your **Min and Max Multiplier**.

Define what you consider a niche journal here, and the tool will take that into account when scoring the projects.

For example: If you want projects that lead to publications in low-volume, high-quality journals, set the Total Docs (3yr) to at most 200 and SJR Quartile to Q1 & Q2.

The final Niche Journal Bonus **formula** will appear dynamically at the bottom.

9

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⌚ Enrichment Schedule

How often the PI enrichment cache is refreshed from public APIs

Enrichment Cycle (days)

14

Last enrichment run: Never

⚠ Unresolved Journals

PubMed abbreviations that the resolution cascade (CrossRef → NLM Catalog → OpenAlex) could not expand into a full title. Helpful for spotting matching gaps.

No unresolved journals — every abbreviation matched at least one tier.

 Save Configuration

 Reset Defaults

Stay up-to-date automatically.

If you leave your data in the browser, the publications and PI information will be updated (enriched) after this set number of days. Last entry is shown for reference.

Check the **log** for any abbreviated journal names that could not be matched to a full journal name, which impacts ability to find journal metrics.

Don't forget to **save** your changes!

The ROI score is calculated using this formula.

$$(S_1 + S_2 + S_3 + S_4) \cdot \sqrt{(2h \div \text{TRACH})}$$

Where the default weighting is:

S₁ Publication Impact = $0.4 \cdot \text{RCR} + 0.4 \cdot \text{NIH\%ile} + 0.2 \cdot \text{Citations}$

S₂ Journal Prestige = $0.25 \cdot \text{SJR-Quartile} + 0.25 \cdot \text{SJR} + 0.25 \cdot \text{h5-Index} + 0.25 \cdot \text{Cit/Doc}$

S₃ Funding Activity = $0.3 \cdot \text{FundingSource} + 0.35 \cdot \text{ActiveGrants} + 0.35 \cdot \text{TotalFunding}$

S₄ Research Output = $0.35 \cdot \text{h-Index} + 0.25 \cdot \text{TotalPubs} + 0.25 \cdot \text{TotalCites} + 0.15 \cdot \text{FacultyRank}$

Notes:

*Niche Bonus and Tier Discounts **can be turned ON** and customized under Admin Config.*

*Weighting **reflects your saved weighting choices** under Admin Config.*

If Niche Bonus and Tier Discounts are turned ON, the formula is:

$$(S_1 + S_2 + S_3 + S_4) \cdot \sqrt{(2h \div \text{TRACH})} \cdot \text{Niche} \cdot [\text{T1: } 1, \text{ T2: } 0.85, \text{ T3: } 0.5]$$

Weight Configuration

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Equal Preset Weights: Scientific Merit **25%** ·
Researcher Track Record **25%** · Institutional
Alignment **25%** · Operational Efficiency **25%**

When to use: The Equal preset distributes weight across all four categories without favoring any single dimension. This is a good neutral starting point when your team has not yet established a prioritization philosophy, or when you want a balanced baseline to compare against other presets.

Total allocation

100.0% ✓



QUICK PRESETS

Equal

Science-heavy

Researcher-heavy

Funding-heavy

Reset equal

Research Prioritizer

EMR Request ROI Scoring

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Science-Heavy Preset Weights: Scientific Merit **38%** · Researcher Track Record **28%** · Institutional Alignment **21%** · Operational Efficiency **13%**

When to use: The Science-Heavy preset prioritizes the quality and impact of the research itself. Use this configuration when your team wants to favor requests tied to high-citation, high-impact publications, or when scientific rigor is the primary criterion for support decisions.

Total allocation

100.0% ✓



● Scientific Merit **38.0%** ● Researcher Track Record **28.0%** ● Institutional Alignment **21.0%** ● Operational Efficiency **13.0%**

QUICK PRESETS

Equal

Science-heavy

Researcher-heavy

Funding-heavy

↺ Reset equal

Research Prioritizer

EMR Request ROI Scoring

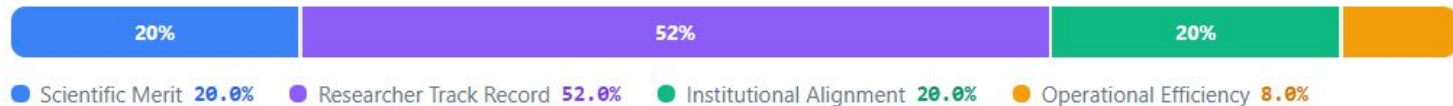
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Researcher-Heavy Preset Weights: Scientific Merit **20%** · Researcher Track Record **52%** · Institutional Alignment **20%** · Operational Efficiency **8%**

When to use: The Researcher-Heavy preset places the most emphasis on the PI's proven history of output. This is ideal when your team prioritizes supporting researchers with a strong, established track record particularly useful in environments where the PI's reputation is a strong signal of project success.

Total allocation

100.0% ✓



QUICK PRESETS

Equal

Science-heavy

Researcher-heavy

Funding-heavy

Reset equal

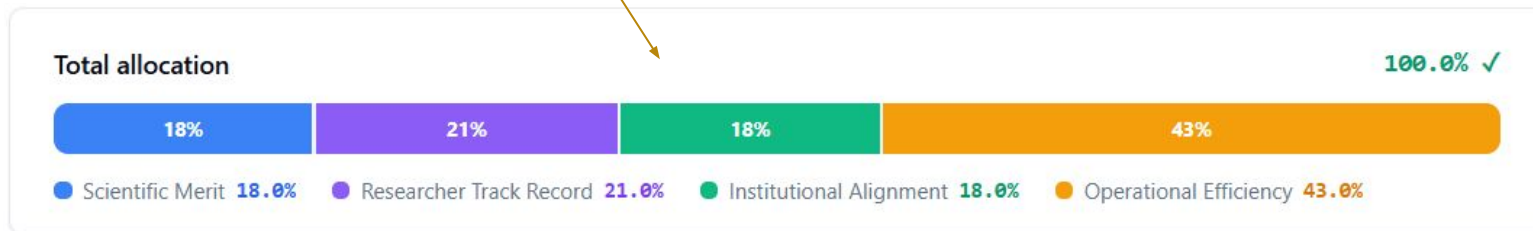
Research Prioritizer

EMR Request ROI Scoring

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Funding-Heavy Preset Weights: Scientific Merit **18%** · Researcher Track Record **21%** · Institutional Alignment **18%** · Operational Efficiency **43%**

When to use: The Funding-Heavy preset emphasizes grant activity and resource efficiency as the primary drivers of prioritization. Choose this configuration when your team needs to align support decisions closely with funding availability and operational capacity.



QUICK PRESETS

Equal

Science-heavy

Researcher-heavy

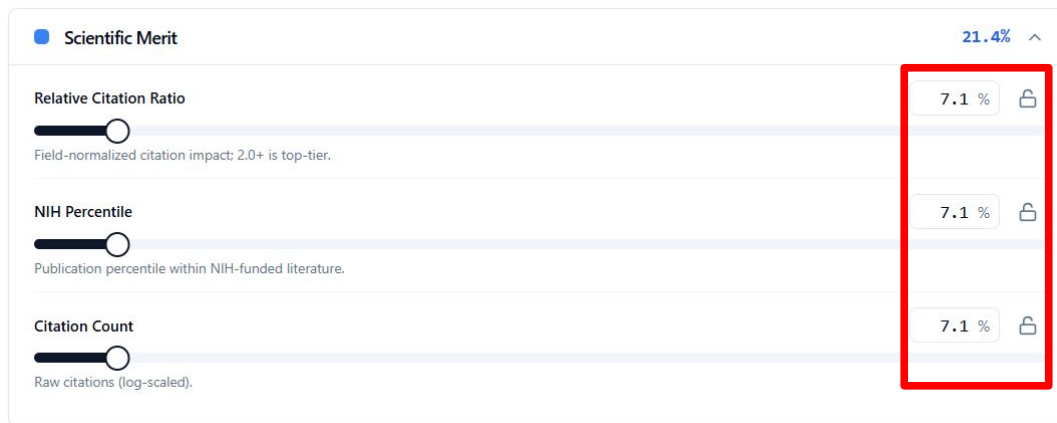
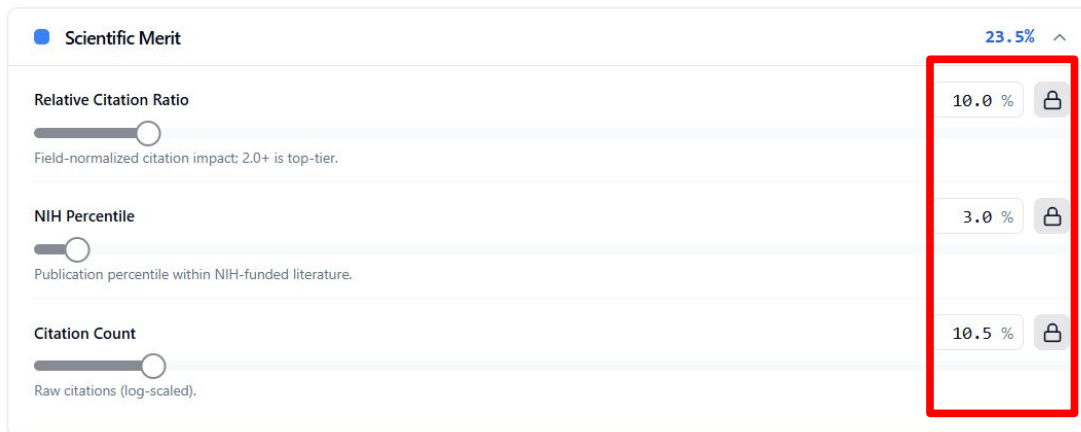
Funding-heavy

Reset equal

Research Prioritizer

EMR Request ROI Scoring

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Beyond the top-level category weights, you can fine-tune scoring within each category using the subweight sliders. For example, within Scientific Merit, you can independently adjust the influence of Relative Citation Ratio, NIH Percentile, and Citation Count. Use the lock icon next to any subweight to fix it in place. When a subweight is locked, adjusting the other sliders will redistribute weight only among the unlocked items, keeping your locked value stable. Thoughtful subweight configuration can meaningfully shift how projects rank, so it is worth revisiting these settings as your team develops a clearer sense of what drives research success at your institution.

Research Prioritizer

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Once you have configured your desired weighting, you can save it as a named preset for future use. Enter a name in the text field and click **Save Weights**. Your preset will appear in the **Saved Presets** list, where it will be marked as Active.

Test 1



Save weights



Saved presets

Click a preset to load its values into the editor and make it active for scoring.

● Default

Click to apply

4/22/2026



● Test 1

✓ Active

4/24/2026



You can return to any saved preset at any time by clicking it. This will load its values into the editor and apply it to scoring immediately. The Default preset is always available as a fallback.

Dashboard

Ranked Request Queue

38 of 38 requests ⓘ Queue updated: 4/25/2026, 8:28:56 PM

🔍 PubMed Publication Search No API key — slower rate limit

Queries the PubMed API using **extracted keywords** from each project title to match DIVE projects to published academic output.

🔍 Search PI, department, title...

🔍 Search PubMed (38)

👤 PI Grant Enrichment 34 pending

👤 Enrich PIs (34)

📊 ROI Scoring Engine ⓘ

📊 Score Requests

How ROI is calculated

Preset: Default

$$\text{ROI} = (0.1 \cdot \text{RCR} + 0.1 \cdot \text{NIH\%ile} + 0.05 \cdot \text{Citations} + 0.06 \cdot \text{SJR-Quartile} + 0.06 \cdot \text{SJR} + 0.06 \cdot \text{h5-Index} + 0.06 \cdot \text{Cit/Doc} + 0.08 \cdot \text{FundingSource} + 0.09 \cdot \text{ActiveGrants} + 0.09 \cdot \text{TotalFunding} + 0.09 \cdot \text{h-Index} + 0.06 \cdot \text{TotalPubs} + 0.06 \cdot \text{TotalCites} + 0.04 \cdot \text{FacultyRank}) \cdot \sqrt{2h + \text{TRAC}_h}$$

GROUPED BY CATEGORY (SAME SUM)

S_1 Publication Impact = $0.1 \cdot \text{RCR} + 0.1 \cdot \text{NIH\%ile} + 0.05 \cdot \text{Citations}$

S_2 Journal Prestige = $0.06 \cdot \text{SJR-Quartile} + 0.06 \cdot \text{SJR} + 0.06 \cdot \text{h5-Index} + 0.06 \cdot \text{Cit/Doc}$

S_3 Funding Activity = $0.08 \cdot \text{FundingSource} + 0.09 \cdot \text{ActiveGrants} + 0.09 \cdot \text{TotalFunding}$

S_4 Research Output = $0.09 \cdot \text{h-Index} + 0.06 \cdot \text{TotalPubs} + 0.06 \cdot \text{TotalCites} + 0.04 \cdot \text{FacultyRank}$

Each sub-metric is normalized to 0–100 before being multiplied by its weight. Missing sub-metrics simply contribute 0.

Edit coefficients on the [Weight Config](#) page — this formula updates live.

Tier discount is **disabled** — all tiers contribute at 1.0× (toggle on Admin Config).

Ranked Request Queue

38 of 38 requests Queue updated: 4/25/2026, 8:28:56 PM

Search PI, department, title...

PubMed Publication Search

No API key — slower rate limit

Search PubMed (38)

PI Grant Enrichment

34 pending

Enrich PIs (34)

Pulls grant funding data, active grants, and researcher metadata from NIH RePORTER and WHO RePORT for each PI — populating the funding-side variables in the ROI formula.

ROI Scoring Engine

Score Requests

How ROI is calculated

Preset: Default

$$ROI = (0.1 \cdot RCR + 0.1 \cdot NIH\%ile + 0.05 \cdot Citations + 0.06 \cdot SJR\text{-}Quartile + 0.06 \cdot SJR + 0.06 \cdot h5\text{-}Index + 0.06 \cdot Cit/Doc + 0.08 \cdot FundingSource + 0.09 \cdot ActiveGrants + 0.09 \cdot TotalFunding + 0.09 \cdot h\text{-}Index + 0.06 \cdot TotalPubs + 0.06 \cdot TotalCites + 0.04 \cdot FacultyRank) \cdot \sqrt{2h + TRAC_h}$$

GROUPED BY CATEGORY (SAME SUM)

- S_1 Publication Impact = $0.1 \cdot RCR + 0.1 \cdot NIH\%ile + 0.05 \cdot Citations$
- S_2 Journal Prestige = $0.06 \cdot SJR\text{-}Quartile + 0.06 \cdot SJR + 0.06 \cdot h5\text{-}Index + 0.06 \cdot Cit/Doc$
- S_3 Funding Activity = $0.08 \cdot FundingSource + 0.09 \cdot ActiveGrants + 0.09 \cdot TotalFunding$
- S_4 Research Output = $0.09 \cdot h\text{-}Index + 0.06 \cdot TotalPubs + 0.06 \cdot TotalCites + 0.04 \cdot FacultyRank$

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Ranked Request Queue

38 of 38 requests Queue updated: 4/25/2026, 8:28:56 PM

Search PI, department, title...

PubMed Publication Search No API key — slower rate limit

Search PubMed (38)

PI Grant Enrichment 34 pending

Enrich PIs (34)

Runs the full ROI scoring engine across all enriched requests turning them into a single ranked score with effort adjustment applied.

ROI Scoring Engine

Score Requests

How ROI is calculated

Preset: Default

$$\text{ROI} = (0.1 \cdot \text{RCR} + 0.1 \cdot \text{NIH\%ile} + 0.05 \cdot \text{Citations} + 0.06 \cdot \text{SJR-Quartile} + 0.06 \cdot \text{SJR} + 0.06 \cdot \text{h5-Index} + 0.06 \cdot \text{Cit/Doc} + 0.08 \cdot \text{FundingSource} + 0.09 \cdot \text{ActiveGrants} + 0.09 \cdot \text{TotalFunding} + 0.09 \cdot \text{h-Index} + 0.06 \cdot \text{TotalPubs} + 0.06 \cdot \text{TotalCites} + 0.04 \cdot \text{FacultyRank}) \cdot \sqrt{2h + \text{TRAC}_h}$$

GROUPED BY CATEGORY (SAME SUM)

S_1 Publication Impact = $0.1 \cdot \text{RCR} + 0.1 \cdot \text{NIH\%ile} + 0.05 \cdot \text{Citations}$

S_2 Journal Prestige = $0.06 \cdot \text{SJR-Quartile} + 0.06 \cdot \text{SJR} + 0.06 \cdot \text{h5-Index} + 0.06 \cdot \text{Cit/Doc}$

S_3 Funding Activity = $0.08 \cdot \text{FundingSource} + 0.09 \cdot \text{ActiveGrants} + 0.09 \cdot \text{TotalFunding}$

S_4 Research Output = $0.09 \cdot \text{h-Index} + 0.06 \cdot \text{TotalPubs} + 0.06 \cdot \text{TotalCites} + 0.04 \cdot \text{FacultyRank}$

Each sub-metric is normalized to 0–100 before being multiplied by its weight. Missing sub-metrics simply contribute 0.

Edit coefficients on the [Weight Config](#) page — this formula updates live.

Tier discount is **disabled** — all tiers contribute at 1.0× (toggle on Admin Config).

Filters:

 **Filters** ☐ Has publications ☐ Has grants ☐ Has follow-on grants ☐ Has IRB

Year requested

From: Any ▾ - To: Any ▾

Status

All ▾

Funding source

Any ▾

Faculty rank

Any ▾

Tier

☐ T1 ☐ T2 ☐ T3

- **Has publications** — Request matched to at least one academic publication via PubMed.
- **Has grants** — PI has grant funding returned from NIH RePORTER or WHO RePORT.
- **Has follow-on grants** — PI secured additional funding following their DIVE engagement.
- **Has IRB** — Request involves an IRB-approved study.

- **Year requested** — Filter by submission date range.
- **Status** — Filter by current position in the DIVE workflow.
- **Funding source** — Filter by grant funding type (e.g. NIH, industry, internal).
- **Faculty rank** — Filter by PI academic rank, which feeds the FacultyRank scoring variable.
- **Tier (T1 / T2 / T3)** — Filter by data completeness at scoring. T1 is fully populated; T2 and T3 have increasing missing data.

Example #1: #551 Promis AYA

Ranked Request Queue

1 of 38 requests ⌚ Queue updated: 4/25/2026, 8:28:56 PM

Filters

☐ Has publications

☐ Has grants

☐ Has follow-on grants

☐ Has IRB

Year requested

From: Any - To: Any

Status

All

Funding source

Any

Faculty rank

Any

Tier

☐ T1 ☐ T2 ☐ T3

#	ROI	Return	Status	Requested	Pubs	Type	Department	Rank	PI	Project Title	Funding
3	33.2	33.2	Closed	12/7/17	7	Identified Data	PHS	Associate Professor	Salsman, John	PROMIS AYA	Federal

Very good ROI

Publications found

Funding source

Example #2: #480 Pediatric Obesity as a Risk Factor for Adverse Safety Events

Ranked Request Queue

1 of 38 requests Queue updated: 4/25/2026, 8:46:53 PM

#	ROI	Return	Status	Requested	Pubs	Type	Department	Rank
9	15.7	15.7	Closed	11/16/17	9	—	Pediatrics	Assistant Professor

Rank	PI	Project Title	Funding
Assistant Professor	Halvorson, Elizabeth	Pediatric Obesity as a Ris...	Institution/Internal/Departmental

PI Profiles

The filters help users quickly find the right PI profiles by narrowing results based on publications, grants, and faculty rank.

PI Profiles

Compare principal investigators by funding, publications, and research output

Search by name or department...

Filters ☒ Has publications ☐ Has follow-on grants

Faculty rank: ☒ Assistant Professor ☒ Associate Professor ☒ Professor

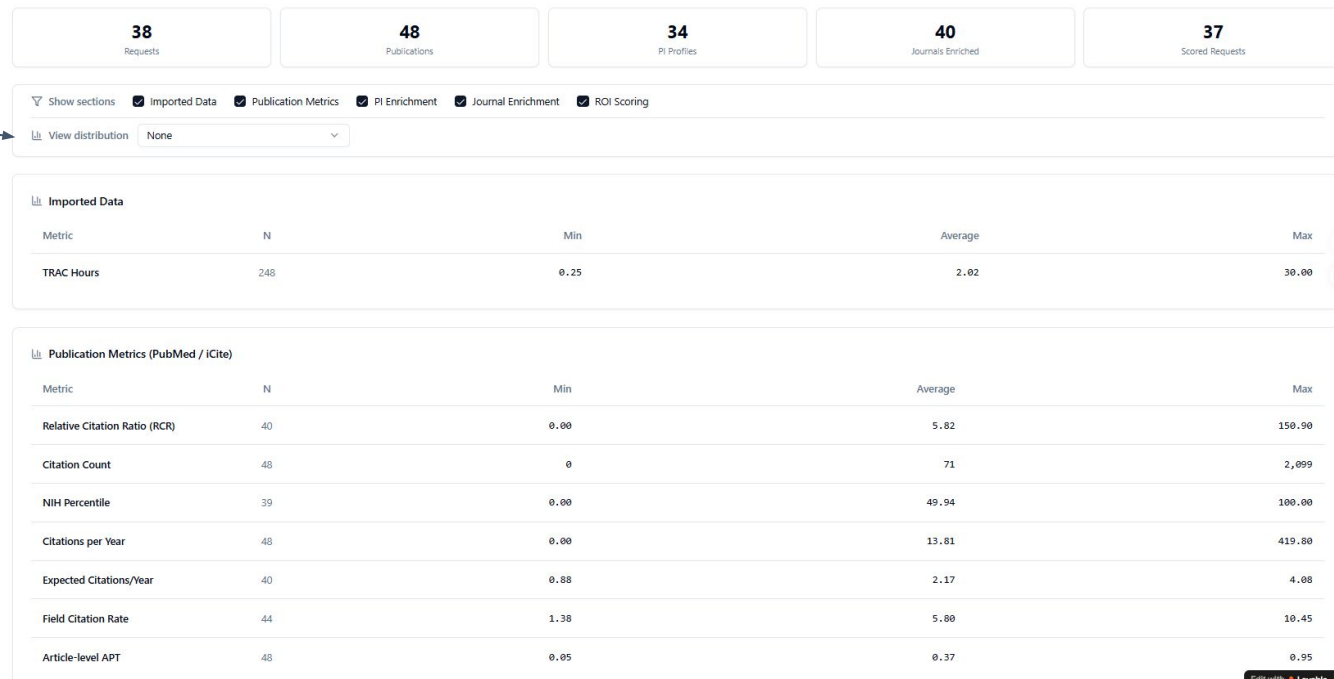
PI Name	Faculty Rank	Requests	Funding T1	Grants	Active	Pubs	Avg RCR	Avg Authors	Dept
Halvorson, Elizabeth	Assistant Professor	1	—	—	—	9	0.80	7.1	Pediatrics
Su, Jing	Assistant Professor	1	—	—	—	7	30.92	428.3	biostatistical sciences, PHS
Whitlow, Christopher	Associate Professor	1	—	—	—	15	1.40	13.0	Radiology
Henderson, Brittany	Assistant Professor	1	—	—	—	1	0.00	7.0	Medicine
Alexander-Miller, Martha	Professor	1	—	—	—	1	0.85	6.0	Microbiology and Immunology
Nunn, Andrew	Assistant Professor	1	—	—	—	1	0.33	8.0	Surgery
Salsman, John	Associate Professor	2	—	—	—	4	1.66	10.8	PHS
Topaloglu, Umit	Associate Professor	2	—	—	—	5	1.13	5.2	Cancer Biology
Herrera, Lourdes	Assistant Professor	1	—	—	—	1	6.16	27.0	Pediatrics-Gastroenterology
Welch, Cherrie	Professor	1 (1 open)	—	—	—	4	2.69	112.0	Pediatrics-Neonatology

Summary Statistics

Lets users choose which metric sections to display, making the summary easier to review.

Summary Statistics

Min, average, and max for all metrics across imported and enriched data



Summarizes publication performance, including citation impact, citation counts, NIH percentile, and citations per year.

Summarizes PI enrichment metrics, including H-index, total citations, publications, grants, and funding

PI Enrichment (NIH RePORTER)

Metric	N	Min	Average	Max
H-Index	10	0	3	7
Total Citations	10	0	343	3,000
Total Publications	10	1	5	15
Grant Count	0	–	–	–
Active Grants	0	–	–	–
Total Grant Funding (\$)	0	–	–	–

Journal enrichment metrics and market context, including SJR, journal H-index, citation quality, market cap, and growth rate

Journal Enrichment (SCImago / Google Scholar)

Metric	N	Min	Average	Max
HS-Index (Google Scholar)	1	399	399	399
HS-Median (Google Scholar)	1	509	509	509
SJR	21	0.79	2.18	11.99
SJR H-Index	21	24	149	368
Total Docs (3yr)	21	122	816	1,709
Citations/Doc (2yr)	21	1.59	4.53	21.71
Market Cap (\$B)	38	6.00	79.32	200.00
Market CAGR (%)	38	5.00	7.33	13.50

ROI scoring results,
including publication
impact, journal prestige,
funding activity, weighted
return, and effort-adjusted
ROI

ROI Scoring				
Metric	N	Min	Average	Max
Publication Impact Score	10	0.00	11.78	20.70
Journal Prestige Score	7	8.40	11.23	16.90
Funding Activity Score	37	2.50	4.26	7.50
Research Output Score	0	-	-	-
Weighted Return Score	37	2.50	9.56	39.60
Effort-Adjusted ROI	37	1.30	9.56	39.60
Effort Multiplier	37	0.52	0.99	1.41

Audit Log

Audit Log

Complete history of manual overrides and scoring actions.

Data Dictionary

Definitions for every column and metric shown across the Dashboard and PI Profiles tables.

DASHBOARD COLUMNS

#	ROI	Return
Row index in the current sorted view.	Effort-Adjusted ROI score: weighted return ÷ effort multiplier, after tier discount.	Weighted Return: sum of 31-54 sub-scores (0-100), pre-effort.
Status	Requested	Pubs
Open or Closed, based on whether the request has a completion date.	Date the REDCap request was submitted (dateOfRequest).	Count of PubMed publications linked to this request.
Type	Department	Rank
Request type from REDCap (e.g., Identified Data, De-identified Data).	Requesting PI's department.	PI faculty rank: Professor, Associate, Assistant, Instructor, or Other.
PI	Project Title	Funding
Principal Investigator name.	Free-text project title from REDCap.	Funding source category (NIH, Foundation, Internal, Industry, etc.).
Tier		
Data confidence tier: T1 = full enrichment, T2 = partial (prony used), T3 = insufficient data.		

PI PROFILE COLUMNS

PI Name	Faculty Rank	Requests
Researcher name (normalized; merged aliases shown as "aka").	Highest faculty rank seen across this PI's requests.	Total number of REDCap requests submitted by this PI.
Funding	Grants	Active
Total NIH RePORTER funding aggregated across all grants, in millions USD.	Total distinct grants on file from NIH RePORTER (deduped by core project number).	Number of currently active (non-expired) NIH grants.
Pubs	Avg RCR	Avg Authors
Total publications linked across all of this PI's requests.	Mean Relative Citation Ratio across this PI's publications.	Mean number of authors per publication for this PI (from PubMed AuthorList).
Dept		
Department associated with this PI in REDCap.		

Defines each dashboard column, including ROI, status, request details, publication count, rank, and funding

Defines each PI profile column, including PI name, faculty rank, requests, grants, publications, RCR, and department

Explains the main scoring components and data tiers used in the ROI model, such as publication impact, researcher track record, journal prestige, funding activity, and data confidence levels

SCORING COMPONENTS & TIERS

S1 — Publication Impact <i>Weighted blend of RCR, NIH Percentile, and Citation Count.</i>	S2 — Researcher Track Record <i>h-index, total publications, total citations, and faculty rank.</i>	S3 — Journal Prestige <i>SJR Quartile, SJR, h5-index, and CIt/Doc 2yr.</i>
S4 — Funding Activity <i>Funding source, active grants, and total funding.</i>	Effort Multiplier <i>$\text{clamp}(\text{anchorHours} + \text{trackHours})^{\wedge} \text{dampening} \cdot \text{Imax}, \text{min}\text{a}[]$. Lower hours → higher multiplier.</i>	Tier 1 (T1) <i>Fully enriched (publications + journal + PI metrics) — no discount applied.</i>
Tier 2 (T2) <i>Partial data; proxy estimates used. Proxy discount applied (if enabled in Admin Config).</i>	Tier 3 (T3) <i>Insufficient data. Insufficient-data discount applied (if enabled in Admin Config).</i>	Niche Bonus <i>Optional multiplier for low-saturation journals matching SCImago criteria (quartile + numeric thresholds).</i>

REFERENCE METRICS

RCR <i>Relative Citation Ratio (NIH iCite): field- and time-normalized citation impact. 1.0 = NIH-funded average.</i>	NIH Percentile <i>Percentile rank of citation rate among NIH-funded papers in the same field/year.</i>	Citation Count <i>Total citations for the publication (from NIH iCite).</i>
Authors <i>Number of authors on a publication (parsed from PubMed AuthorList).</i>	h-index <i>Researcher's h-index — number of publications cited at least h times each.</i>	h5-index <i>Google Scholar 5-year h-index for the journal: articles in past 5 yrs each cited a h5 times.</i>
h5-median <i>Median citation count of the h5 articles — typical impact within that core set.</i>	SJR <i>SCImago Journal Rank: prestige-weighted citations per document over the past 3 years.</i>	SJR Quartile <i>Q1 (top 25%) – Q4 ranking within the journal's SCImago subject category.</i>
CIt/Doc 2yr <i>Average citations per document over the past 2 years (SCImago).</i>	Total Docs (3yr) <i>Articles published by the journal over the past 3 years (SCImago).</i>	

Defines the key reference metrics used in scoring, including RCR, NIH percentile, citation count, H-index, SJR, and journal citation measures

Export

Research Prioritizer

EMR Request ROI Scoring

- Dashboard
- PI Profiles
- Summary Stats
- Weight Config
- Admin Config
- Audit Log
- Upload Data
- Export

Export & Reporting

Download scored data, enrichment results, and audit logs.

Export Filters

Filter by status and date range. Leave empty to include all.

Status: **All** Open Closed

Dates: From date to To date

Comprehensive Report

Multi-sheet .xlsx with ranked requests, PI profiles, journal metrics, publications, audit log, and weight config.

Export Full Report 38 scored - 34 PIs - 61 Journals

Open Requests (Ranked)

Ranked by Effort-Adjusted ROI with subscores, PI metrics, and enrichment data.

Export 4 Open Requests 34 closed excluded

Raw Data Backup

All request data (open + closed) as .xlsx backup

Download All 38 Requests

Data Summary

Open Requests	4	Closed Requests	34
Scored Requests	38	PI Profiles	34
Journal Records	61	Publications	80
Manual Overrides	0	Upload Logs	1

Export a file to save it or use in another tool.

Open requests only can be used as a current prioritization list.

Raw data is a copy of your REDCap and Trac files combined.

Filter your export for custom reports.

All requests with ROI, PI, journal, and publication, grant, and market data can be used as a static file for future predictive modeling, natural language processing, and exploratory statistics.

See how many of **each type** of record are in the tool.

The Power Arrangers have created a **niche tool** designed to be used internally by CTSI's DIVE Team to **identify high-impact health data research** that show high potential to bring funds, prestige, and community impact to the School of Medicine.

The niche ROI scoring tool will allow you to realize **business impact**.

50%

Reduction in Manual Triage

Within 3 months

80%

Team Adoption Rate

As primary workflow tool

99.9%

System Uptime

During business hours

Strategic Value

Increase High-Impact Research Support

Measurably track the proportion of completed requests benefiting federally-funded or high-impact studies

Boost Research Output

Compare publication rates from serviced requests 12 months pre- vs. post-launch

Build Institutional Data Asset

Train ML models in 12-18 months to enable fully predictive, proactive triage

Identify Strategic Research Areas

Discover niche and high-growth medical specialties for competitive advantage

The following **limit** ROI score assessment capabilities.

- 1** Only **historical metrics**; no predictive analytical modeling
- 2** Not all external data extraction is **automated**
- 3** **Missing** values in original dataset; missing private information like institutional, industry, and nonprofit funding

Next year's cohort can further **enhance** ROI assessment capabilities.

- 1 Natural language processing** of given: project titles, inclusion/exclusion criteria and data elements requested
- 2** Analytical modeling to create a **projected ROI score** for incoming requests based on historical data
- 3** Incorporate **public data** other than NIH RePORTER information (see sources marked with a  in appendix).

The tool application will be shared with you through **GitHub**.

- 1** **Make an account** at [GitHub](#). Send Julia your username at reicje25@wfu.edu and she will add you to the private GitHub repository.
- 2** You will **receive Admin rights** and we will not add or change any of the underlying code from now on.
- 3** If you would like to **continue to edit the tool by vibecoding**, we recommend creating a [Lovable.ai](#) account that connects directly to your GitHub account.

APPENDIX

The following are **sources** for each piece of the ROI puzzle,
dynamically updated in the background.

FUNDING

Identified Source

Data Files in repositories

WFU Internal / Departmental

unknown

Nonprofit



National Center for Science and Engineering Statistics and Census Bureau, Nonprofit Research Activities module of the [Annual Business Survey](#).

Industry



Business Enterprise Research and Development (BERD) [Survey 2023](#)



National Center for Science and Engineering Statistics, FFRDC Research and [Development Survey, FY 2024](#).

Federal



[NIH RePORTER](#) API from scraped grants linked to PubMed articles



National Science Foundation - [Awards Simple Search](#) for past and present awards to "Wake Forest University"

[PCORI Portfolio Search](#)

JOURNALS

Identified Source	h5-index	h5-median	SJR	h-index	Total Docs (3yr)
Purpose	<i>A journal has an h5-index of n if it published n articles in the past 5 years that each received at least n citations.</i>	<i>The median citation count of those n articles, giving a sense of the typical impact within that core set.</i>	<i>Avg weighted citations received by the documents published in the journal in the three previous years. (Q1 top tier)</i>	<i>Journal's number of articles (h) that have received at least h citations over the whole period.</i>	<i>Journal's published articles (all types) in 2022, 2023, and 2024.</i>
Data Files in repositories	 <u>Google Scholar</u>* * Only top 100 healthcare journals		<u>SCImago</u>		



Note: **Clarivate** denied our request for an API key. You may have more luck as an established institution, where you will be able to find similar but informational journal metrics.

CITATIONS

Identified Source	<i>RCR</i>	<i>NIH Percentile</i>	<i>Authors</i>
Purpose	<i>field- and time-normalized citation rate</i>	<i>RCR comparison to other public publications</i>	<i>author count</i>
Data Files in repositories	<i>iCite</i>		

GLOBAL MARKET VALUATION

Identified
Source

*Global World
Spending*

*Global Market
Size Estimation*

Data Files in
public
repositories

[WHO World
RePORT](#)
(global biomedical
research grant
investments)
2023

[Global
Observatory of
Health R&D](#)
(by WHO)



Pitchbook
(manual search)
(WFU Subscription)